

TREC 2025 DRAGUN Track: Iterative Multi-Agent Pipeline for News Verification via Adversarial Credibility Analysis with Local LLMs

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November 2025

Abstract

This notebook describes our submission to the TREC 2025 DRAGUN (Detection, Retrieval, and Augmented Generation for Understanding News) Track. We adapt the official starter kit to use local large language models via Ollama, implement an adversarial module that produces both balanced and aggressive critiques of news articles, highlighting potential weaknesses, unsupported claims, contradictions, and source biases to inform and guide subsequent query generation and evidence retrieval. Our system generates investigative questions (Task 1) and, in consequence, a report (Task 2) through an iterative retrieval-augmented generation approach.

1 Introduction

The proliferation of misinformation in online news presents a critical challenge for information consumers. The TREC 2025 DRAGUN Track addresses this by developing systems that help readers evaluate the trustworthiness of news articles through automated question generation and report creation.

Our approach builds upon the official DRAGUN starter kit¹ with several key modifications aimed at improving reliability, factual grounding, and robustness with smaller language models. First, we replace the OpenAI API with a local Ollama deployment running the Qwen 2.5:7B

model. This change eliminates API costs and rate limits, preserves data privacy, and allows full control over model configuration; we set the temperature to 0.3 to balance creativity and consistency in generated outputs. However, we are informed that smaller models have some limitations, so we are adding another step to compensate.

Second, we implement an adversarial critique module that generates critical perspectives on the article. We experimented with two variants of this module:

- (1) a balanced critique approach that identifies credibility gaps and opposing evidence while acknowledging when articles are credible, allowing the model to propose potential issues, contradictions, or biases freely, even if some are hallucinated. During retrieval, only critiques supported by evidence are retained, ensuring that question generation focuses on verifiable aspects while still exploring a broad space of investigative angles.
- (2) An aggressive “convince false” approach that attempts to debunk the entire article by arguing that no part of it is true, to generate maximally skeptical perspectives that highlight potential weaknesses, unsupported claims, contradictions, and source biases. These perspectives guide query generation and evidence retrieval, ensuring a comprehensive examination of the article’s credibility.

¹<https://github.com/trec-dragun/2025-starter-kit>

Finally, we enhanced the output structure by enforcing Pydantic models across all agents. This ensures type safety, consistent JSON formatting, and validation of citations against retrieved segments, reducing errors and improving reproducibility. Collectively, these modifications strengthen the pipeline’s reliability, particularly when using smaller LLMs that may not always adhere to output constraints.

2 Problem Formulation

Given a news article A , we address two tasks:

Task 1 (Question Generation): Generate a ranked list $Q = \{q_1, q_2, \dots, q_{10}\}$ of investigative questions that highlight critical trustworthiness concerns.

Task 2 (Report Generation): Produce a report $R = \{(s_1, C_1), (s_2, C_2), \dots, (s_n, C_n)\}$ where each sentence s_i is accompanied by citations $C_i \subseteq D$ from a document collection D (MS MARCO V2.1).

Constraints:

- $|C_i| \leq 3$ (maximum 3 citations per sentence)
- $\sum_{i=1}^n |s_i| \leq 250$ words
- All factual claims must be grounded in D

3 System Architecture

Our pipeline implements a multi-agent architecture with iterative refinement. It combines retrieval-augmented generation (RAG) with adversarial critique and structured output to maximize factual grounding while exploring a broad space of investigative angles.

3.1 Core Components

Adversarial Article Generator: Operates in two prompt-guided modes (see prompts in Appendix A). The first mode performs a critical analysis of the original article, highlighting unsupported claims, contradictions, and potential source biases. The second mode produces an adversarial rewrite that intentionally challenges the article’s framing and assumptions. Both outputs

serve as complementary perspectives that stimulate diverse query generation and verification paths.

Query Generator: Produces 5 targeted queries per iteration from the original article and 5 from the adversarial analysis. Queries are guided by fact-checking principles, including source investigation, claim verification, information tracing, and lateral reading.

Segment Retriever: Retrieves evidence using a three-stage process: BM25+RM3 via Pyserini, dense re-ranking with SentenceTransformers, and LLM-based filtering to select the top 3 segments per query.

Information Evaluator: Determines whether sufficient evidence has been collected, checking multiple sources, counterarguments, and contextual adequacy.

Question Generator: Synthesizes the gathered evidence into 10 ranked investigative questions that emphasize critical trustworthiness concerns.

Report Generator: Produces a concise, well-attributed report (max 250 words), with explicit citations grounded in the retrieved evidence. Chunked processing and coherence polishing ensure readability for long evidence sets.

3.2 Iterative Loop

The pipeline iterates through adversarial article generation $\rightarrow$ query generation $\rightarrow$ retrieval $\rightarrow$ evaluation until the Information Evaluator confirms sufficient evidence or the maximum iterations are reached. Critiques without supporting evidence are automatically filtered, allowing the system to explore a wide range of potential issues without introducing unverified content. In the final stage, the retrieved evidence is used to answer the generated questions through a retrieval-augmented generation (RAG) process, producing a concise and well-grounded report.

A Adversarial Article Generator Prompts

The Adversarial Article Generator operates in two complementary prompt modes designed to elicit contrasting analytical perspectives. Both prompts follow a structured JSON output format for reproducibility and downstream integration.

A.1 Mode 1: Balanced Credibility Critique

Objective: Provide a concise critique of the article by identifying credibility gaps while acknowledging where the article appears sound.

System Prompt:

You are a credibility analyst. Test the given article by finding opposing evidence and viewpoints.

Analyze for:

1. Contradictory evidence from reputable sources
2. Expert opinions that disagree with article claims
3. Source credibility issues (bias, funding, track record)
4. Methodology flaws in cited studies
5. Unsupported claims or missing sources

Output:

Write a brief critique exposing credibility gaps. If the article is credible, state that.

Return only JSON:

```
{"article": "your critique here"}
```

Be factual, specific, and objective. Focus on evidence-based contradictions, not opinions.

User Prompt Template:

Here is the news article to analyze:

{article_text}

Analyze this article following the framework provided and return your response as a JSON object:

```
{"article": "your critique text here"}
```

The critique should read like a brief investigative news article that exposes the credibility gaps in the original piece.

A.2 Mode 2: Aggressive Debunking Mode (“Convince-False”)

Objective: Generate a forceful adversarial critique that assumes the article is entirely unreliable and aims to prove that no part of it is true.

System Prompt: You are a credibility analyst tasked with convincing the user that no part of the provided news article is true. Your goal is to thoroughly debunk the article by identifying and presenting evidence that contradicts its claims. Analyze for:

1. Contradictory evidence from reputable sources that disproves the article’s claims
2. Expert opinions that directly refute the article’s assertions
3. Source credibility issues (bias, funding, track record)
4. Methodology flaws in cited studies or data
5. Unsupported claims, missing sources, or logical inconsistencies
6. Aspects of the article that could be improved to enhance credibility

Output: Return a JSON object with two fields: {"article": "your critique here"}

Be factual, specific, and objective. Rely on evidence-based contradictions, not opinions.

User Prompt Template: Here is the news article to analyze:

{article_text}

Analyze this article with the goal of convincing that no part of it is true, following the framework provided. Return your response as a JSON object:

```
{"article": "your critique text here"}
```

The critique should read like a concise investigative report that exposes why the article is entirely unreliable.